

- **Lab Title:** Lab 4 Steganography and Hidden Data Detection
- **Assignment Title:** Steganography and Hidden Data Detection
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- **Student ID:** 2025/FWSD/11509
- **Course Name:** SBT-DF202 Computer and Digital Forensics
- **Instructor Name:** Aminu Idris
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- **Version:** Version 2.2.3

Platform: Kali Linux or Ubuntu Forensic Workstation / VM

Lab Overview

By completing this lab, students will demonstrate their ability to explain the difference between steganography and encryption, describe insertion and substitution-based hiding techniques including least significant bits (LSB), prepare a carrier image and secret evidence file, use Steghide to embed and extract concealed data, calculate and compare cryptographic hashes, perform a controlled password dictionary test using StegSeek, and document all activities in a repeatable forensic report

Introduction

Steganography is a technique used to conceal information within an ordinary-looking digital file, such as an image, audio file, video, or document, in a way that makes the existence of the hidden information difficult to detect. Unlike encryption, which makes the content unreadable without a key, steganography focuses on hiding the presence of the information itself. This makes it an important area of study in digital forensics and cybersecurity because hidden data can potentially be used to conceal confidential information, evidence, or malicious content.

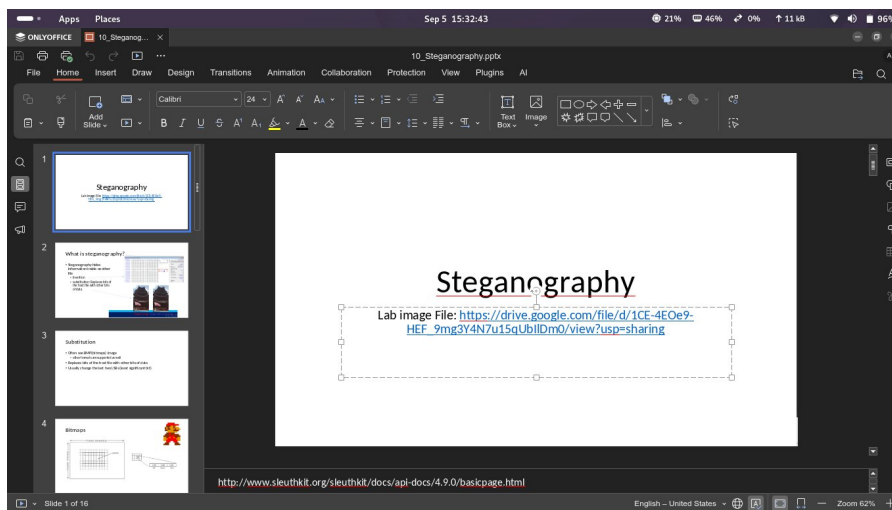
This practical laboratory focuses on the application of steganography and hidden-data detection techniques using BMP images. The laboratory involved creating and embedding secret information into a carrier image using Steghide, extracting the hidden information, verifying file integrity using SHA-256 hashing and file comparison, and testing password recovery using StegSeek and controlled password wordlists. The practical also included an independent student-created steganography exercise to demonstrate the complete process of embedding, extraction, verification, and password recovery.

The purpose of this laboratory is to develop practical forensic skills in identifying, recovering, and validating concealed information while maintaining the integrity and reliability of digital evidence.

Steganography Lab Carrier Image

Before beginning practical work, I reviewed the Module 8 lessons and study material on Steganography and Hidden Data Analysis, covering hidden information, substitution techniques, least significant bits (LSB), bitmap images, Steghide, extraction, and password recovery, as required by the lab's instructor briefing. I created a dedicated working directory with separate subfolders for evidence, secret files, stego images, extracted files, hashes, logs, screenshots, wordlists, and reports.

I downloaded the authorized lab carrier image, tower_original_image_for_lab.bmp, from the course-provided source. file confirmed it is a genuine PC bitmap, Windows 3.x format, 1365×2265 pixels, 24-bit color depth, with a pixel data offset of 54 bytes into the file this uncompressed, high-bit-depth format is exactly the kind of carrier Steghide requires, since LSB steganography needs raw, uncompressed pixel data to embed hidden bits into without a lossy compression algorithm destroying them afterward.



```
azad@localhost: ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
Unpacking gdown (6.1.0+dfsg-1)...
Setting up python3-filelock (3.32.2-1)...
Setting up gdown (6.1.0+dfsg-1)...
Processing triggers for kali-menu (2026.3.2)...
ls: cannot access 'tower_original_image_for_lab.bmp': No such file or directory
tower_original_image_for_lab.bmp: cannot open 'tower_original_image_for_lab.bmp' (No such file or directory)

[azad@localhost] ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
$ gdown --id 1CE-4EOe9-HEF_9mg3Y4N7u15qUblDm0 -O tower_original_image_for_lab.bmp
ls -lh tower_original_image_for_lab.bmp
file tower_original_image_for_lab.bmp
usage: gdown [-h] [-v] [-O OUTPUT] [-q] [--proxy PROXY] [--speed SPEED]
              [--no-cookies] [--no-check-certificate] [--continue] [--folder]
              [--json] [--format FORMAT] [--user-agent USER_AGENT]
              url_or_id
gdown: error: unrecognized arguments: --id
ls: cannot access 'tower_original_image_for_lab.bmp': No such file or directory
tower_original_image_for_lab.bmp: cannot open 'tower_original_image_for_lab.bmp' (No such file or directory)

[azad@localhost] ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
$ ls
tower_original_image_for_lab.bmp

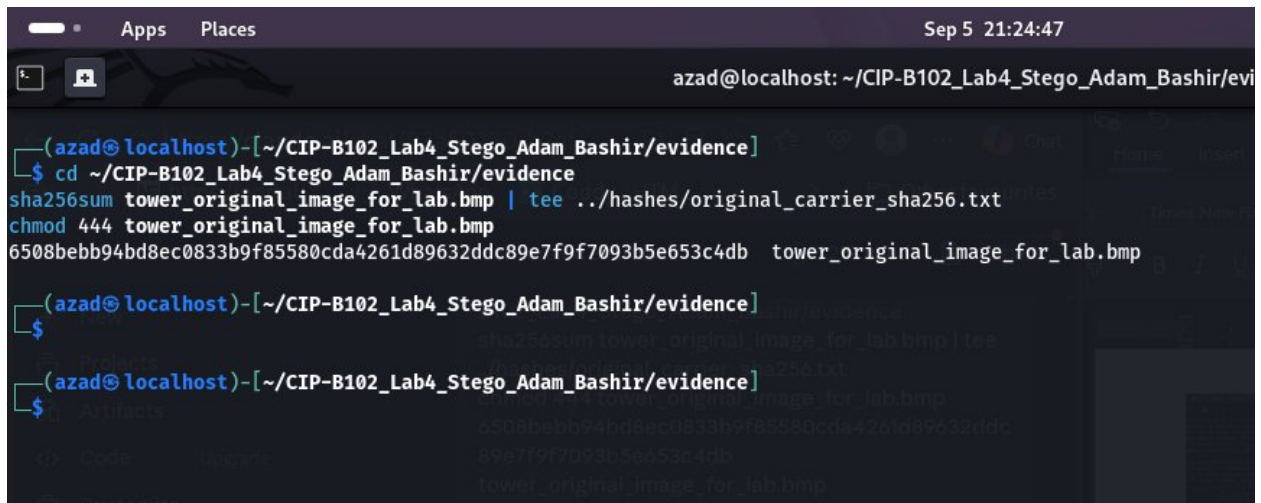
[azad@localhost] ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
$ cd ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
rm -f tower_original_image_for_lab.bmp
gdown 1CE-4EOe9-HEF_9mg3Y4N7u15qUblDm0 -O tower_original_image_for_lab.bmp
ls -lh tower_original_image_for_lab.bmp
file tower_original_image_for_lab.bmp
Download...
From: https://drive.google.com/uc?id=1CE-4EOe9-HEF_9mg3Y4N7u15qUblDm0
To: /home/azad/CIP-B102_Lab4_Stego_Adam_Bashir/evidence/tower_original_image_for_lab.bmp
100%|#####| 9.28M/9.28M [03:13<00:00, 47.9kB/s]
-rw-r--r-- 1 azad azad 8.9M Dec  9 2025 tower_original_image_for_lab.bmp
tower_original_image_for_lab.bmp: PC bitmap, Windows 3.x format, 1365 x 2265 x 24, resolution 2834 x 2834 px/m, cbSize 9277494, bits offset 54

[azad@localhost] ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
$
```

Part A Prepare and Preserve the Evidence

Baseline SHA-256 of the original carrier image:

6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db. The file was then set to read-only (chmod 444) to preserve it and prevent accidental overwriting during the embedding exercise, per the lab's explicit requirement.



```
(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence]
$ cd ~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence
sha256sum tower_original_image_for_lab.bmp | tee ../hashes/original_carrier_sha256.txt
chmod 444 tower_original_image_for_lab.bmp
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db  tower_original_image_for_lab.bmp

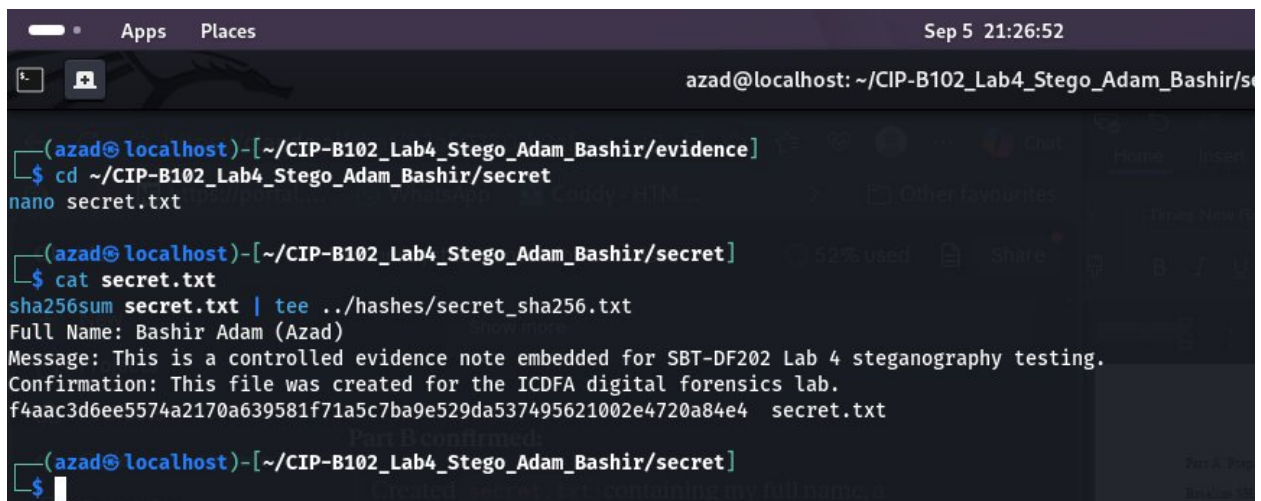
(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence]
$

(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir/evidence]
$
```

Part B Create the Secret Evidence File

Created secret.txt containing my full name, a controlled evidence message, and confirmation the file was made for the ICDFA lab, exactly as required. SHA-256:

f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4.



```
(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir/secret]
$ cd ~/CIP-B102_Lab4_Stego_Adam_Bashir/secret
nano secret.txt

(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir/secret]
$ cat secret.txt
Full Name: Bashir Adam (Azad)
Message: This is a controlled evidence note embedded for SBT-DF202 Lab 4 steganography testing.
Confirmation: This file was created for the ICDFA digital forensics lab.
f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4  secret.txt

(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir/secret]
$
```

Part C Embed the Hidden File with Steghide

Steghide embedded secret.txt into the carrier image using password 1234, producing tower_stego_lab4.bmp (8.9M same size as the original, since LSB substitution doesn't add data, only alters existing bit values). SHA-256:

e526205dc2daaf9ce0c104935e00bfb2789f6c1e19d71733c0e067d43af51220 already visibly different from the original carrier's hash (6508beb...), confirming the embedding changed the file's content at the byte level despite the identical file size.

```
azad@localhost: ~/CIP-B102_Lab4_Stego_Adam_Bashir
$ sudo apt install steghide -y
cd ~/CIP-B102_Lab4_Stego_Adam_Bashir
steghide embed -ef secret/secret.txt -cf evidence/tower_original_image_for_lab.bmp -sf stego/tower_stego_lab4.bmp -p 1234
ls -lh stego/tower_stego_lab4.bmp
sha256sum stego/tower_stego_lab4.bmp | tee hashes/stego_sha256.txt
[sudo] password for azad:
Installing:
  steghide

Installing dependencies:
  libmcrypt4  libmhash2

Suggested packages:
  libmcrypt-dev  mcrypt

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 438
  Download size: 319 kB
  Space needed: 966 kB / 348 GB available

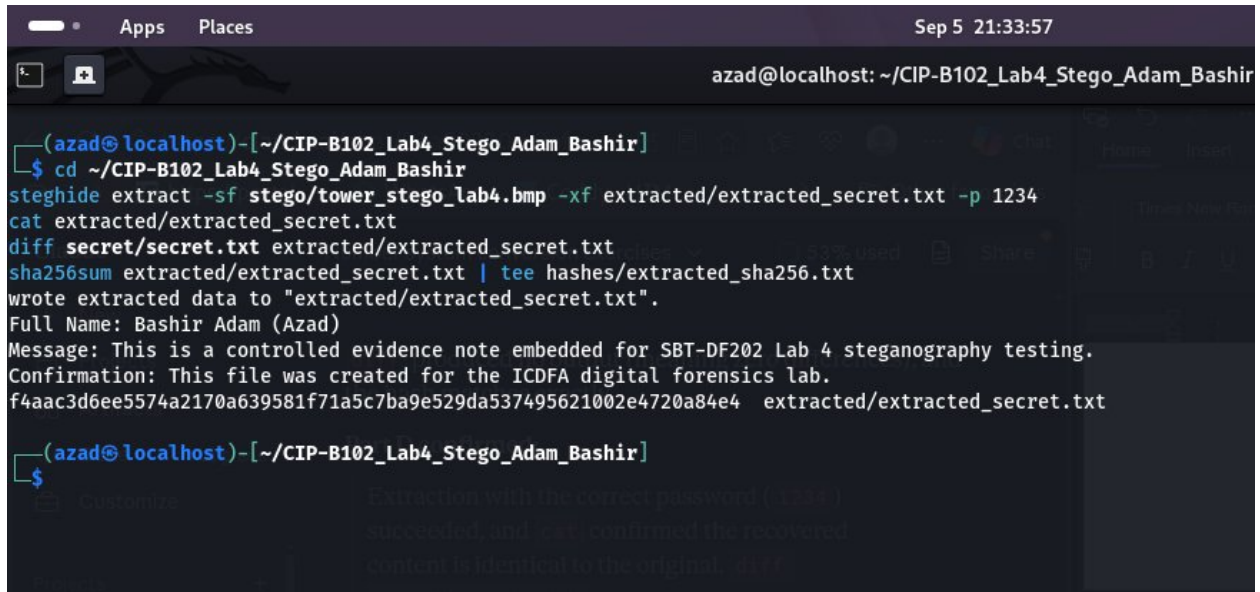
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libmcrypt4 amd64 2.5.8-8+b2 [71.8 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libmhash2 amd64 0.9.9-9-11+b1 [93.3 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 steghide amd64 0.5.1+git20240220-2 [154 kB]
Fetched 319 kB in 8s (42.4 kB/s)
Selecting previously unselected package libmcrypt4:amd64.
(Reading database... 316804 files and directories currently installed.)
Preparing to unpack ./libmcrypt4_2.5.8-8+b2_amd64.deb...
Unpacking libmcrypt4:amd64 (2.5.8-8+b2)...
Selecting previously unselected package libmhash2:amd64.
Preparing to unpack ./libmhash2_0.9.9-9-11+b1_amd64.deb...
Unpacking libmhash2:amd64 (0.9.9-9-11+b1)...
Selecting previously unselected package steghide.
Preparing to unpack ./steghide_0.5.1+git20240220-2_amd64.deb...
Unpacking steghide (0.5.1+git20240220-2)...
Setting up libmhash2:amd64 (0.9.9-9-11+b1)...
Setting up libmcrypt4:amd64 (2.5.8-8+b2)...
Setting up steghide (0.5.1+git20240220-2)...

Setting up libmhash2:amd64 (0.9.9-9-11+b1)...
Setting up libmcrypt4:amd64 (2.5.8-8+b2)...
Setting up steghide (0.5.1+git20240220-2)...
Processing triggers for libc-bin (2.42-17)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.3.2)...
embedding "secret/secret.txt" in "evidence/tower_original_image_for_lab.bmp"... done
writing stego file "stego/tower_stego_lab4.bmp"... done
-rw-rw-r-- 1 azad azad 8.9M Sep  5 21:28 stego/tower_stego_lab4.bmp
e526205dc2daaf9ce0c104935e00bfb2789f6c1e19d71733c0e067d43af51220  stego/tower_stego_lab4.bmp

(azad@localhost)~-[~/CIP-B102_Lab4_Stego_Adam_Bashir]
$
```

Part D Extract and Verify the Hidden Evidence

Extraction with the correct password (1234) succeeded, and cat confirmed the recovered content is identical to the original. diff secret/secret.txt extracted/extracted_secret.txt returned no output at all, which in diff means the two files are completely identical no lines added, removed, or changed. sha256sum confirmed this cryptographically: f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4, exactly matching the original secret.txt hash from Part B. This proves Steghide's extraction is fully lossless the hidden file survived embedding and extraction with zero data corruption.



```
(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir]
$ cd ~/CIP-B102_Lab4_Stego_Adam_Bashir
steghide extract -sf stego/tower_stego_lab4.bmp -xf extracted/extracted_secret.txt -p 1234
cat extracted/extracted_secret.txt
diff secret/secret.txt extracted/extracted_secret.txt
sha256sum extracted/extracted_secret.txt | tee hashes/extracted_sha256.txt
wrote extracted data to "extracted/extracted_secret.txt".
Full Name: Bashir Adam (Azad)
Message: This is a controlled evidence note embedded for SBT-DF202 Lab 4 steganography testing.
Confirmation: This file was created for the ICDFa digital forensics lab.
f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4  extracted/extracted_secret.txt

(azad@localhost)-[~/CIP-B102_Lab4_Stego_Adam_Bashir]
$
```

Part E Compare Original and Stego Images

Property	Original BMP	Stego BMP
File size	8.9 MB	8.9 MB
SHA-256	6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db	e526205dc2daaf9ce0c104935e00bfb2789f6c1e19d71733c0e067d43af51220
BMP header	Present	Present
Visual appearance	Normal	Normal
Hidden data	No	Yes

Despite the file size being byte-for-byte identical and the visible header being unchanged, the SHA-256 hash is completely different between the two files. This is the core forensic lesson of this comparison: SHA-256 is sensitive to a change in even a single bit anywhere in the file, so even though a human eye (or an image viewer) cannot visually detect any difference between the original and stego images, the underlying binary content specifically, the least significant bits of many individual pixel values has been altered throughout the image to encode the hidden message. This demonstrates why hash comparison, not visual inspection, is the reliable method for detecting whether an image has been tampered with or used to conceal data.

```
[~] Apps Places Sep 5 22:42:18 azad@localhost: ~/CIP-B102_Lab4_Stego_Adam_Bashir  
[azad@localhost]~/CIP-B102_Lab4_Stego_Adam_Bashir$ cd ~/CIP-B102_Lab4_Stego_Adam_Bashir  
ls -lh evidence/tower_original_image_for_lab.bmp stego/tower_stego_lab4.bmp  
sha256sum evidence/tower_original_image_for_lab.bmp stego/tower_stego_lab4.bmp  
xxd evidence/tower_original_image_for_lab.bmp | head -5  
xxd stego/tower_stego_lab4.bmp | head -5  
strings stego/tower_stego_lab4.bmp | grep -i '\ICDFA\|secret\|Bashir'  
-r--r--r-- 1 azad azad 8.9M Dec 5 2025 evidence/tower_original_image_for_lab.bmp  
-rw-rw-r-- 1 azad azad 8.9M Sep 5 21:28 stego/tower_stego_lab4.bmp  
6508bbe9b4bd8e083b3bf8f5580cdca4261d89632ddc89ef9ff7093b5e653cadb evidence/tower_original_image_for_lab.bmp  
e526205dc2daaf9cce0104953e00bf2789f6fc1e9d71733ce0e67d43af51220 stego/tower_stego_lab4.bmp  
00000000: 424d 3690 8d00 0000 0000 3600 0000 2800 BM6.....6...(  
00000010: 0000 5505 0000 d908 0000 0100 1800 0000 ..U.....  
00000020: 0000 0000 0000 120b 0000 120b 0000 0000 .....  
00000030: 0000 0000 0000 1e22 3520 2338 1e20 381f ..... "5 #8. 8.  
00000040: 203a 2425 3f28 2945 2729 482a 2b4d 2a2c :%$(E)'H**+M*,  
00000000: 424d 3690 8d00 0000 0000 3600 0000 2800 BM6.....6...(  
00000010: 0000 5505 0000 d908 0000 0100 1800 0000 ..U.....  
00000020: 0000 0000 0000 120b 0000 120b 0000 0000 .....  
00000030: 0000 0000 0000 1e22 3520 2338 1e20 381f ..... "5 #8. 8.  
00000040: 203a 2425 3f28 2945 2729 482a 2b4d 2a2c :%$(E)'H**+M*,  
{!uopekfgfabicdfabz[e'ac'd_"]^"\[\ZYZ'\gbccZUVNIJB>8345015/03-.2,-/)0**+1+,1+,1+,0+0**+0+-3.-3-,1+,3-./01+,0+-3-/5/04/. /7126012,-3,-3.-3-/./3.-2,-2,-1,+0+  
+/0+0**+)0+*-2,-4/-2,-1,+1+,1+,0+*)*(.(0)+*+1+,2,-4/. /3.-3.-4./2,-2,-2,-2,-1,+1+,1+,1+,0+*1+,2,-1,+0+*)*(0)+*+2,-2,-1,+2,-2,-2,-3.-4./3.-2,-3+2**3.+3+  
4/4/,1,)0+(2-x+1[,])/* /*'0+(1,j2>+1,+1,+2,-3.-2,-2,-2,-3.-2,-2,-2,-0+0+2-*2+2+1,j2+2-*2+3.+3+. /3.,+1)0+0(1,j2+4,/3.-2,-2,-2,-3.-4./5/04/. /4./3+  
-2,-2,-2,-2,-2,-1+,3-50/4/,3.-2,-1,+0+2,-2,-2,-1,+3+2k+1))3+2,-2,-2,-2,-2,-3.-3.-3.-3.-3.-3.-3.-1*-1*-0+*.().(0)+2,-2,-3.-4./2,-1,)*+1/3.-1+,.().0+*/)*(  
0)+2,-3.-3.-2,-2,-3.-0+*+14/,3-3.-2,-3.-3.-2,-3.-4./4-03/,2,-2,-1,+1+,1+,1+,1+,1+,2,-2,-0+*.().(/)*)*(.).().).0+*1,1,1,1,2-/w/16135024/14/14/14/16135/45/43-22,13-2,  
2,12,14,33-23-2/+30+45-7;/3=q8BFHA<EA(E=BA<7/72;-2-64,971<7172;9294,36051+00*/0*/0+/100-/302A>qOLNXUNJGI>9;/57968968746857:796351.01.01.0+((**))-#,-0-/0-/3020/-..+++,  
)')%'0../.-,*-0,1-),(.1,54.90+5,(410:1/;64q<9HGDTMK'KH'LlbnkeMjdnLIomJRpmTRoQmOpLNokPMVZsyWt_uO/r-uYgZ}2h~!  
  
[azad@localhost]~/CIP-B102_Lab4_Stego_Adam_Bashir$
```

[illegible]

Part F Controlled Password Dictionary Test

A controlled wordlist containing the known password 1234 was created and tested against the stego image using StegSeek. StegSeek successfully identified the passphrase 1234 and extracted the embedded file, originally named secret.txt. The recovered file was compared with the original using diff, which produced no differences. SHA-256 hashing produced the same hash for both files (f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4). This confirms successful password recovery and integrity-preserving extraction.

```
azad@localhost: ~/CIP-B102_Lab4_Stego_Adam_Bashir
$ cd ~/CIP-B102_Lab4_Stego_Adam_Bashir && printf "1234\n" > wordlists/lab4_wordlist.txt && cat wordlists/lab4_wordlist.txt
1234

$ which stegseek
stegseek not found

$ stegseek stego/tower_stego_lab4.bmp wordlists/lab4_wordlist.txt
Command 'stegseek' not found, but can be installed with:
sudo apt install stegseek
Do you want to install it? (N/y)y
sudo apt install stegseek
[sudo] password for azad:
Installing: stegseek
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 438
  Download size: 130 kB
  Space needed: 375 kB / 348 GB available

Get:1 http://http.kali.org/kali-rolling/main amd64 stegseek amd64 0.6+git20210910.ff677b9-2 [130 kB]
Fetched 130 kB in 2s (63.0 kB/s)
Selecting previously unselected package stegseek.
(Reading database... 316837 files and directories currently installed.)
Preparing to unpack ./stegseek_0.6+git20210910.ff677b9-2_amd64.deb...
Unpacking stegseek (0.6+git20210910.ff677b9-2)...
Setting up stegseek (0.6+git20210910.ff677b9-2)...
Processing triggers for kali-menu (2026.3.2)...

$ stegseek stego/tower_stego_lab4.bmp wordlists/lab4_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "1234"
[i] Original filename: "secret.txt".

$ diff secret/secret.txt stego/tower_stego_lab4.bmp.out && sha256sum secret/secret.txt stego/tower_stego_lab4.bmp.out
diff: stego/tower_stego_lab4.bmp.out: No such file or directory

$ ls -lh tower_stego_lab4.bmp.out && diff secret/secret.txt tower_stego_lab4.bmp.out && sha256sum secret/secret.txt tower_stego_lab4.bmp.out
-rw-rw-r-- 1 azad azad 199 Sep  5 23:06 tower_stego_lab4.bmp.out
f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4  secret/secret.txt
f4aac3d6ee5574a2170a639581f71a5c7ba9e529da537495621002e4720a84e4  tower_stego_lab4.bmp.out

$
```


Part G Independent Student Task

An independent forensic steganography test was performed using a student-created hidden note. The note was embedded into the BMP carrier using Steghide with the password CyberLab2026!. Successful extraction was confirmed using Steghide, and the recovered file produced the same SHA-256 hash as the original (f4d4d1f83ce7db7c19d9f40d271a33f19e1de821ed096f1f7c09b49cb56c2124). A custom password wordlist was subsequently tested with StegSeek. StegSeek successfully recovered the password CyberLab2026! and extracted the embedded file. The extracted file also produced the same SHA-256 hash, confirming successful recovery and data integrity.

```
azad@localhost: ~/CIP-B102_Lab4_Stego_Adam_Bashir
$ nano secret/Adam_Bashir_hidden_note.txt
$ cat secret/Adam_Bashir_hidden_note.txt
Steganography is the practice of hiding information inside another file, such as an image, audio file, video, or document, so that the hidden information is not immediately noticeable.
Steganography is important in digital forensics because criminals or other users may conceal evidence, confidential information, or malicious data inside ordinary-looking files. A forensic examiner can use hashing, file analysis, metadata examination, and specialized steganography tools to detect and recover hidden information.
One common image steganography concept is LSB (Least Significant Bit) substitution. It hides information by modifying the least significant bits of pixel values. These small changes may not be visible to the human eye, while still allowing data to be stored inside the image.
$ sha256sum secret/Adam_Bashir_hidden_note.txt | tee hashes/Adam_Bashir_hidden_note_original.sha256
f4d4d1f83ce7db7c19d9f40d271a33f19e1de821ed096f1f7c09b49cb56c2124 secret/Adam_Bashir_hidden_note.txt
$ steghide embed -cf stego/tower_stego_lab4.student.bmp -ef secret/Adam_Bashir_hidden_note.txt -p "CyberLab2026!"
dquote> y Take a screenshot of the successful embedding message.
dquote>
```

```
azad@localhost: ~/CIP-B102_Lab4_Stego_Adam_Bashir
$ mkdir -p extracted/student
$ steghide extract -sf stego/tower_stego_lab4.bmp -p 'CyberLab2026!' -xf extracted/student/Adam_Bashir_hidden_note_extracted.txt
wrote extracted data to "extracted/student/Adam_Bashir_hidden_note_extracted.txt".
$ diff secret/Adam_Bashir_hidden_note.txt extracted/student/Adam_Bashir_hidden_note_extracted.txt 66 sha256sum secret/Adam_Bashir_hidden_note.txt extracted/student/Adam_Bashir_hidden_note_extracted.txt
f4d4d1f83ce7db7c19d9f40d271a33f19e1de821ed096f1f7c09b49cb56c2124 secret/Adam_Bashir_hidden_note.txt
f4d4d1f83ce7db7c19d9f40d271a33f19e1de821ed096f1f7c09b49cb56c2124 extracted/student/Adam_Bashir_hidden_note_extracted.txt
$ printf "password\n123456\nCyberLab2026!\nadmin\n" > wordlists/student_wordlist.txt 66 cat wordlists/student_wordlist.txt
zsh: event not found: \nadmin\n
$ printf 'password\n123456\nCyberLab2026!\nadmin\n' > wordlists/student_wordlist.txt 66 cat wordlists/student_wordlist.txt
password
123456
CyberLab2026!
admin
$ stegseek stego/tower_stego_lab4.bmp wordlists/student_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek
[i] Found passphrase: "CyberLab2026!"
[i] Original filename: "Adam_Bashir_hidden_note.txt".
[i] Extracting to "tower_stego_lab4.bmp.out".
[i] The file "tower_stego_lab4.bmp.out" does already exist. overwrite ? (y/n)
y
$ sha256sum tower_stego_lab4.bmp.out extracted/student/Adam_Bashir_hidden_note_extracted.txt 66 diff tower_stego_lab4.bmp.out secret/Adam_Bashir_hidden_note.txt
f4d4d1f83ce7db7c19d9f40d271a33f19e1de821ed096f1f7c09b49cb56c2124 tower_stego_lab4.bmp.out
f4d4d1f83ce7db7c19d9f40d271a33f19e1de821ed096f1f7c09b49cb56c2124 extracted/student/Adam_Bashir_hidden_note_extracted.txt
```

Conclusion

The laboratory successfully demonstrated the practical application of steganography and hidden-data detection techniques in a digital forensic investigation. Secret information was successfully embedded into a BMP carrier image using Steghide and subsequently recovered using the correct passphrase. The recovered files were verified against their original versions using the `diff` command and SHA-256 hashing, with matching hashes confirming that the extracted data remained unchanged.

The controlled password dictionary test using StegSeek successfully identified the embedded password, demonstrating how password-based steganographic containers can be subjected to dictionary-based recovery techniques when an appropriate wordlist is available. The independent student exercise further demonstrated the complete forensic workflow by creating a custom hidden note, embedding it with a separate password, extracting it, and recovering the password using a custom wordlist.

Overall, the practical strengthened the understanding of steganography, LSB/substitution concepts, password-protected hidden data, forensic extraction, hashing, and evidence validation. The results demonstrate that digital forensic investigators can combine specialized steganography tools with hashing and file-comparison techniques to detect, recover, and authenticate concealed digital information while preserving evidence integrity.